



**EFY GROUP**  
YOURS SINCE 1969

# Facing problems in getting EFY Magazine in your area? If so, let us know!



Electronics For You magazine is available at all top news-stands/outlets in India. But if you are facing problem in getting EFY magazine in your area, please call Akhilesh Yadav on +91-8800295612 or drop an email at [efycirc@efy.in](mailto:efycirc@efy.in). We will help you locate the nearest news-stand or arrange a copy at your doorstep.

**electronics**  
**FOR YOU**

South Asia's Most Popular Electronics Magazine

Those who wish to distribute EFY Group publications may also call Akhilesh Yadav on +91-8800295612 or write at [efycirc@efy.in](mailto:efycirc@efy.in)

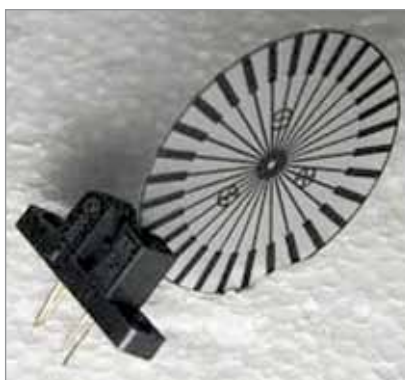


Fig. 5: Photo interrupter



Fig. 6: Slotted disc

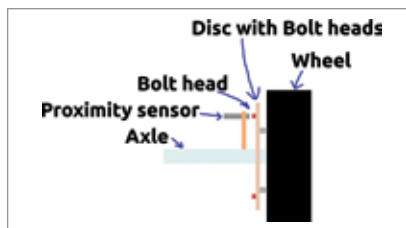


Fig. 7: Proximity

## PARTS LIST

### Semiconductors:

Board1 - Arduino Uno  
D1 - 1N4007 rectifier diode

Resistors (all 1/4-watt,  $\pm 5\%$  carbon), unless stated otherwise:

R1 - 270-ohm  
R2 - 10-kilo-ohm  
VR1 - 10-kilo-ohm potmeter

### Miscellaneous:

S1 - Push-to-on switch  
LCD1 - 16x2 LCD

that this sensor can detect. It has high reliability and long functional life because of the absence of mechanical parts and lack of physical contact between sensor and the sensed objects.

## Construction and testing

An actual-size single-side PCB layout shown in Fig. 3 can be used for the circuit. Its component layout is shown in Fig. 4. Assemble the circuit on PCB and connect 12V and Gnd across CON1 and proximity sensor across CON2. Switch S1 is used to stop/start the circuit. After assembling the circuit enclose it in a suitable enclosure.

The program is written in Arduino programming language (sketch). Arduino IDE is used to compile and upload the program to the Arduino board. Select the correct board from Board→Tools menu in Arduino IDE, select COM port and upload the program through the standard USB port in computer.

Please note the following:

- Most cabs (especially autos) have 40.6cm (16-inch) dia tyres, which means their circumference is approximately 128cm (0.00128km). So, with each rotation of the wheel the distance travelled is 0.00128km.

- In this project, the cost per km is set at ₹9, but it can be changed in the program.

- More than one metal target can be used to get better resolution of the distance travelled. During testing, five targets were used.

- A pull-up or pull-down resistor may be required at output of the sensor, depending on the type of sensor used.

- A proximity sensor with nominal range of 5 to 10mm can be used.

- An opto-slotted IR-photodiode pair with slotted disc (see Figs 5 and 6) can be used for sensing the rotation in place of the proximity sensor.

- The disc should be fixed at the inner side of the wheel as shown in Fig. 7.

EFY

**EFY Note**  
The source code of this project is available for free download at [source.efymag.com](http://source.efymag.com)



A. Samiuddin, a circuit designer, is B.Tech in electrical and electronics engineering. His interests include LED lighting, power electronics, microcontrollers, and Arduino programming